Information Security

Muhammad Bilal, Ph.D.

Ph.D. Computer Science and Technology, Zhejiang University, China

MS Computer Science, NTU Faisalabad Pakistan

B.Sc (Hons) Computer Science, UET Lahore Pakistan

Email:- dr.bilal@numl.edu.pk

National University of Modern Language

Motivation


إِنَّ اللَّهَ مَعَ الصَّابِرِينَ
"Indeed Allah is with those who patiently endure."
[Qur'an, 8:46]

Information Security Model (CIA)

- These three letters stand for **confidentiality, integrity, and availability**, otherwise known as the CIA triad.
- The CIA triad is a **widely used information security model** that can guide an organization's efforts and policies aimed at keeping its data secure.

CIA Triad – The Objectives of Security



CIA Triad – The Objectives of Security

Confidentiality (C)

- Assuring information will be kept secret, with access limited to appropriate (authorized) persons

Integrity (I)

- Assuring information will not be accidentally or maliciously altered or destroyed

Availability (A)

- Assuring information and communications services will be ready for use when expected

CIA Triad – The Objectives of Security

- **Confidentiality**
  - **Data confidentiality**: Assures that confidential information is not disclosed to unauthorized individuals.
  - **Privacy**: Assures that individual control or influence what information may be collected and stored.

CIA Triad – The Objectives of Security

- **Integrity**
  - **Data integrity**: assures that information and programs are changed only in a specified and authorized manner.
  - **System integrity**: Assures that a system performs its operations in unimpaired manner.
- **Availability**: assure that systems works promptly and service is not denied to authorized users.

Other Concepts to Complete Security Picture

- **Authenticity**: the property of being genuine and being able to be verified and trusted; confident in the validity of a transmission, a message, or its originator
- **Accountability**: generates the requirement for actions of an entity to be traced uniquely to that individual to support non-repudiation, deference, fault isolation, etc.

Note:- Authentication proves who you are, and accountability records what you did

Levels of Security Breach Impact

- **Low**: the loss will have a limited impact, e.g., a degradation in mission or minor damage or minor financial loss or minor harm
- **Moderate**: the loss has a serious effect, e.g., significance degradation on mission or significant harm to individuals but no loss of life or threatening injuries
- **High**: the loss has severe or catastrophic adverse effect on operations, organizational assets or on individuals (e.g., loss of life)

Jazak Allah Khair!